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EE 869 – Linear Systems Theory

Fall 2026

Lectures 4 – 6

Mathematical Preliminaries

Linear Independence: The n-dimensional vectors $\{v_1, \ldots, v_m\}$ are linearly dependent if there are m scalars $\alpha_1, \ldots, \alpha_m$ such that

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0$$

They are linearly independent if

$$\sum_{i=1}^{m} \alpha_i v_i = 0 \;\Leftrightarrow\; \alpha_i = 0 \text{ for } i = 1, \ldots, m$$

$\{v_1, \ldots, v_m\}$ are linearly independent if and only if

$$\text{rank}[v_1, \ldots, v_m] = m$$

Let A be $n \times n$. If $\text{rank}(A) = r < n$, then there are $(n - r)$ linearly independent vectors $v_1, \ldots, v_{n-r}$ such that

$$Av_i = 0, \quad \text{for } i = 1, \ldots, n - r$$

A square matrix is nonsingular if and only if all its columns are linearly independent

Eigenvalues and Eigenvectors

For an $n \times n$ matrix A, the eigenvalues and eigenvectors are defined by

$$Av_i = \lambda_i v_i$$

The matrix has n eigenvalues $\lambda_1, \ldots, \lambda_n$ with corresponding eigenvectors $v_1, \ldots, v_n$. The eigenvalues are the roots of the characteristic equation

$$\det(sI - A) = 0$$

If the eigenvalues are distinct, then the eigenvectors are linearly independent

A and $P^{-1}AP$ have the same eigenvalues

Assume A has distinct eigenvalues $\lambda_1, \ldots, \lambda_n$. Then, the eigenvectors $v_1, \ldots, v_n$ are linearly independent

$$P = [v_1, v_2, \ldots, v_n]$$

P is nonsingular

$$\begin{aligned} AP &= A[v_1, v_2, \ldots, v_n] = [Av_1, Av_2, \ldots, Av_n] \\ &= [\lambda_1 v_1, \lambda_2 v_2, \ldots, \lambda_n v_n] \\ &= [v_1, v_2, \ldots, v_n] \begin{bmatrix} \lambda_1 & & & \\ & \cdot & & \\ & & \cdot & \\ & & & \lambda_n \end{bmatrix} \end{aligned}$$

Let

$$\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_n^T \end{bmatrix}$$

$$AP = P\Lambda$$

$$\Lambda = P^{-1}AP$$

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{bmatrix}$$

$$e^{At} = Pe^{\Lambda t}P^{-1} = [v_1, v_2, \ldots, v_n] \begin{bmatrix} e^{\lambda_1 t} & & & \\ & \cdot & & \\ & & \cdot & \\ & & & e^{\lambda_n t} \end{bmatrix} \begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_n^T \end{bmatrix}$$

$$e^{At} = [v_1, v_2, \ldots, v_n] \begin{bmatrix} e^{\lambda_1 t} w_1^T \\ e^{\lambda_2 t} w_2^T \\ \vdots \\ e^{\lambda_n t} w_n^T \end{bmatrix}$$

$$e^{At} = \sum_{i=1}^{n} e^{\lambda_i t} v_i w_i^T$$

Modal Form

Modal Decomposition:

$$e^{At} = \sum_{i=1}^{n} e^{\lambda_i t} v_i w_i^T$$

λ_i, v_i, and w_i may be complex (in conjugate pairs) but e^{At} is real. $e^{\lambda_i t}$ are called the exponential modes of the system

$$\begin{aligned} x(t) &= e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau \\ &= \sum_{i=1}^{n} e^{\lambda_i t} v_i w_i^T x(0) + \sum_{i=1}^{n} \int_0^t e^{\lambda_i (t-\tau)} v_i w_i^T B u(\tau)\,d\tau \end{aligned}$$

w_i^T are called the left eigenvectors of A

$$AP = P\Lambda$$

$$P^{-1}A = \Lambda P^{-1}$$

$$\begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_n^T \end{bmatrix} A = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_n^T \end{bmatrix}$$

$$w_i^T A = \lambda_i w_i^T, \quad \text{for } i = 1, \ldots, n$$

What if A has multiple eigenvalues?

Example

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Eigenvalues:

$$\lambda_1 = \lambda_2 = 1, \lambda_3 = 2$$

Eigenvectors:

$$Av = \lambda v \;\Leftrightarrow\; (A - \lambda I)v = 0$$

$$A - \lambda_1 I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{rank}(A - \lambda_1 I) = 1$$

We can find two linearly independent vectors v_1 and v_2 such that

$$Av_1 = \lambda_1 v_1 \quad \text{and} \quad Av_2 = \lambda_1 v_2$$

Find v_3 such that $Av_3 = \lambda_3 v_3$ and take

$$P = [v_1, v_2, v_3] \quad \Rightarrow \quad P^{-1}AP = \Lambda$$

Example

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

Eigenvalues:

$$\lambda_1 = \lambda_2 = 1, \lambda_3 = 2$$

$$A - \lambda_1 I = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathrm{rank}(A - \lambda_1 I) = 2$$

There is only one linearly independent eigenvector

A is not diagonalizable

Find v_1 such that

$$Av_1 = \lambda_1 v_1 \quad \rightarrow \quad v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Find v_2 (generalized eigenvector) such that

$$Av_2 = \lambda_1 v_2 + v_1 \quad \rightarrow \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Find v_3 such that

$$Av_3 = \lambda_3 v_3$$

It can be shown that $\{v_1, v_2, v_3\}$ are linearly independent

$$\begin{aligned} Av_1 &= \lambda_1 v_1 \\ Av_2 &= \lambda_1 v_2 + v_1 \\ Av_3 &= \lambda_3 v_3 \end{aligned}$$

$$A[v_1, v_2, v_3] = [v_1, v_2, v_3] \left[\begin{array}{cc|c} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ \hline 0 & 0 & \lambda_3 \end{array}\right]$$

In general, a square matrix can be transformed into the form

$$J = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & . & \\ & & & . \\ & & & J_r \end{bmatrix}$$

$$J_i = \lambda_i, \quad \text{or } J_i = \begin{bmatrix} \lambda_i & 1 \\ 0 & \lambda_i \end{bmatrix}, \quad \text{or } J_i = \begin{bmatrix} \lambda_i & 1 & 0 \\ 0 & \lambda_i & 1 \\ 0 & 0 & \lambda_i \end{bmatrix}$$

$$e^{Jt} = \begin{bmatrix} e^{J_1 t} & & & \\ & e^{J_2 t} & & \\ & & \ddots & \\ & & & e^{J_r t} \end{bmatrix}$$

$$J_i = \lambda_i I + N_i$$

$$e^{J_i t} = e^{\lambda_i t} e^{N_i t}$$

$$N_i = 0 \quad \rightarrow \quad e^{N_i t} = 1$$

$$N_i = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \rightarrow \quad e^{N_i t} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

$$N_i = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \rightarrow \quad e^{N_i t} = \begin{bmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$$

The most general representation of e^{At} is

$$e^{At} = \sum_{i=1}^{r} \sum_{k=1}^{m_i} W_{ik} \frac{t^{k-1}}{(k-1)!} e^{\lambda_i t}$$

where m_i is the order of the Jordan block J_i

Stability

Internal Stability

Internal stability deals with the boundedness and asymptotic behavior (as $t \to \infty$) of the solutions of

$$\dot{x} = Ax$$

The solution of $\dot{x} = Ax$ is given by

$$x(t) = e^{At}x(0)$$

Definition: The system $\dot{x} = Ax$ is stable if

$$\left\| e^{At} \right\| \leq \gamma, \quad \forall\, t \geq 0, \quad \gamma > 0$$

and asymptotically (or exponentially) stable if

$$\left\| e^{At} \right\| \leq \gamma e^{-\lambda t}, \quad \forall\, t \geq 0, \quad \gamma > 0, \ \lambda > 0$$

If A is diagonalizable

$$e^{At} = \sum_{i=1}^{n} e^{\lambda_i t} v_i w_i^T$$

The behavior depends on $R_e[\lambda_i]$

$$R_e[\lambda_i] < 0 \quad \Rightarrow \quad \left| e^{\lambda_i t} \right| \leq e^{-\lambda t}, \ \lambda > 0$$

$$R_e[\lambda_i] = 0 \quad \Rightarrow \quad \left| e^{\lambda_i t} \right| = 1$$

$$R_e[\lambda_i] > 0 \quad \Rightarrow \quad e^{\lambda_i t} \text{ is unbounded}$$

In general

$$e^{At} = \sum_{i=1}^{r} \sum_{k=1}^{m_i} W_{ik} \frac{t^{k-1}}{(k-1)!} e^{\lambda_i t}$$

$$R_e[\lambda_i] > 0 \quad \Rightarrow \quad t^{k-1} e^{\lambda_i t} \text{ is unbounded}$$

$$R_e[\lambda_i] < 0 \quad \Rightarrow \quad \left| t^{k-1} e^{\lambda_i t} \right| \leq \gamma e^{-\lambda t}, \ \gamma > 0, \ \lambda > 0$$

$$R_e[\lambda_i] = 0 \text{ and } k = 1 \quad \Rightarrow \quad e^{\lambda_i t} \text{ is bounded}$$

$$R_e[\lambda_i] = 0 \text{ and } k \geq 2 \quad \Rightarrow \quad t^{k-1} e^{\lambda_i t} \text{ is unbounded}$$

When will the dimension of the Jordan block J_i be higher than one?

$$(A - \lambda_i I)v_i = 0$$

Let q_i be the algebraic multiplicity of λ_i. If

$$\text{rank}(A - \lambda_i I) = n - q_i$$

then there are q_i linearly independent eigenvectors associated with the eigenvalue λ_i

If this is true for every eigenvalue that has multiplicity higher than one, then we can find n linearly independent eigenvectors and A is diagonalizable

$$\text{If} \quad \text{rank}(A - \lambda_i I) > n - q_i$$

for any eigenvalue with multiplicity higher than one, then A is not diagonalizable

Theorem: The system $\dot{x} = Ax$ is

- Stable if and only if

$$R_e[\lambda_i] \leq 0, \quad \text{for } i = 1, 2, \ldots, n$$

and for every eigenvalue with $R_e[\lambda_i] = 0$ and algebraic multiplicity $q_i \geq 2$,

$$\text{rank}(A - \lambda_i I) = n - q_i$$

- Asymptotically (or exponentially) stable if

$$R_e[\lambda_i] < 0, \quad \text{for } i = 1, 2, \ldots, n$$

Example: Study the stability of $\dot{x} = Ax$, where

$$A = \begin{bmatrix} -1 & 0 & 3 & 1 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

$$\text{Eigenvalues}: \quad -1, -2, -3, -3$$

The system is asymptotically stable

Example: Study the stability of $\dot{x} = Ax$, where

$$A = \begin{bmatrix} -1 & 0 & 3 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Eigenvalues}: \quad -1, 1, 0, 0$$

The system is unstable

Example: Consider the series and parallel connections of two identical systems, each represented by

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad y = \begin{bmatrix} 1 & 0 \end{bmatrix} x$$

or

$$H(s) = \frac{1}{s^2 + 1}$$

$$A_s = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix}, \quad A_p = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$\text{Eigenvalues}: \quad \pm j, \ \pm j$$

Series Case:

$$\lambda_1 = j, \quad q_1 = 2, \quad n - q_1 = 2$$

$$A_s - \lambda_1 I = \begin{bmatrix} -j & 1 & 0 & 0 \\ -1 & -j & 0 & 0 \\ 0 & 0 & -j & 1 \\ 1 & 0 & -1 & -j \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -j & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & -j \end{bmatrix}$$

$$\text{rank}(A_s - \lambda_1 I) = 3$$

The system is unstable

Parallel Case:

$$\lambda_1 = j, \quad q_1 = 2, \quad n - q_1 = 2$$

$$A_p - \lambda_1 I = \begin{bmatrix} -j & 1 & 0 & 0 \\ -1 & -j & 0 & 0 \\ 0 & 0 & -j & 1 \\ 0 & 0 & -1 & -j \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -j & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -j \end{bmatrix}$$

$$\text{rank}(A_p - \lambda_1 I) = 2$$

The system is stable

What is the effect of state transformations on stability?

$$A \rightarrow \boxed{x = Pz} \rightarrow P^{-1}AP$$

$$Av = \lambda v$$

$$P^{-1}Av = \lambda P^{-1}v$$

$$(P^{-1}AP)(P^{-1}v) = \lambda P^{-1}v$$

A and $P^{-1}AP$ have the same eigenvalues. If v_i is an eigenvector of A, then $P^{-1}v_i$ is an eigenvector of $P^{-1}AP$

$$\text{rank}\,(P^{-1}AP - \lambda I) = \text{rank}\,[P^{-1}(A - \lambda I)P] = \text{rank}\,(A - \lambda I)$$

Internal stability is invariant under state transformations

What is the relationship between asymptotic and BIBO stability?

The system $\dot{x} = Ax$ is asymptotically stable if all the eigenvalues of A have negative real parts.

The system $H(s) = C(sI - A)^{-1}B + D$ is BIBO stable if all poles of all elements of $H(s)$ have negative real parts

The poles of $H(s)$ are eigenvalues of A

Asymptotic stability $\Rightarrow$ BIBO stability

What about the opposite implication?

Some eigenvalues of A may not appear as poles of $H(s)$. If such eigenvalues have nonnegative real parts, then we could have a situation where the system is BIBO stable but not asymptotically stable

Example: Suppose we want to stabilize an unstable system described by

$$G_p(s) = \frac{1}{s-1}$$

Consider a cascade compensator

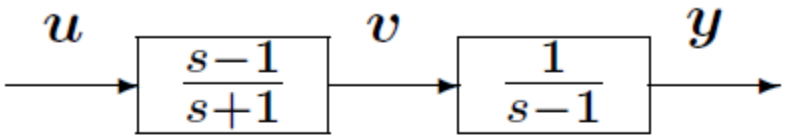
$$G_c(s) = \frac{s-1}{s+1}$$

so that

$$G_p(s)G_c(s) = \frac{1}{s-1} \cdot \frac{s-1}{s+1} = \frac{1}{s+1}$$

The system is BIBO stable. Is it asymptotically stable?

Find a state model of the system



$$G_p(s) = \frac{1}{s-1} \quad \rightarrow \quad \dot{x}_1 = x_1 + v, \;\; y = x_1$$

$$G_c(s) = \frac{s-1}{s+1} \quad \rightarrow \quad \dot{x}_2 = -x_2 + u, \;\; v = -2x_2 + u$$

$$\dot{x}_1 = x_1 - 2x_2 + u$$

$$\dot{x} = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$A = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$$

$$\text{Eigenvalues} : 1, -1$$

The system is not asymptotically stable. The eigenvalue $+1$ is called a hidden mode. Unstable hidden modes are not acceptable because they can be excited by initial conditions or disturbances

For example

$$x_1(0) = \alpha,\ x_2(0) = 0,\ u(t) \equiv 0 \ \Rightarrow\ x_2(t) \equiv 0$$

$$\Rightarrow\ x_1(t) = \alpha e^t$$

Discrete-Time Systems

A linear time-invariant discrete-time system is represented by the state model

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

k is the time variable

$$k = 0, 1, 2, \ldots$$

Sampled-data systems:

Suppose the linear time-invariant continuous-time system

$$\begin{aligned} \dot{x} &= A_c x + B_c u \\ y &= C_c x + D_c u \end{aligned}$$

is preceded by a zero-order hold, i.e., the control input u is kept constant over the sampling period

$$u(t) = u_k, \quad \text{for } t_k \leq t < t_{k+1}$$

$$t_{k+1} - t_k = T, \quad \forall\, k \geq 0$$

Find a discrete-time model that describes the behavior of the system at the sampling points.

Recall

$$x(t) = e^{A_c t} x(0) + \int_0^t e^{A_c (t-\tau)} B_c u(\tau)\, d\tau$$

$$x(t_{k+1}) = e^{A_c (t_{k+1} - t_k)} x(t_k) + \int_{t_k}^{t_{k+1}} e^{A_c (t_{k+1} - \tau)} B_c u(\tau)\, d\tau$$

$$x(t_{k+1}) = e^{A_c T} x(t_k) + \int_{t_k}^{t_{k+1}} e^{A_c (t_{k+1} - \tau)} B_c\, d\tau\, u_k$$

$$\sigma = t_{k+1} - \tau$$

$$\int_{t_k}^{t_{k+1}} e^{A_c (t_{k+1} - \tau)} B_c\, d\tau = -\int_T^0 e^{A_c \sigma} B_c\, d\sigma = \int_0^T e^{A_c \sigma} B_c\, d\sigma$$

$$x(t_{k+1}) = e^{A_c T} x(t_k) + \int_0^T e^{A_c \sigma} B_c \, d\sigma \, u_k$$

Define

$$A = e^{A_c T}, \quad B = \int_0^T e^{A_c \sigma} B_c \, d\sigma$$

$$x(k+1) = Ax(k) + Bu(k)$$

$$y(t_k) = C_c x(t_k) + D_c u(t_k)$$

$$y(k) = Cx(k) + Du(k)$$

where

$$C = C_c, \quad D = D_c$$

In summary, given a continuous-time system $\{A_c, B_c, C_c, D_c\}$ driven by a zero-order hold of sampling period T, the discrete-time equivalent is given by $\{A, B, C, D\}$, where

$$A = e^{A_c T}, \quad B = \int_0^T e^{A_c \sigma} B_c \, d\sigma$$

$$C = C_c, \quad D = D_c$$

Example: Find the discrete-time equivalent of

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x$$

$$e^{A_c t} = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix} \Rightarrow A = \begin{bmatrix} \cos T & \sin T \\ -\sin T & \cos T \end{bmatrix}$$

$$e^{A_c t} B_c = \begin{bmatrix} \sin t \\ \cos t \end{bmatrix} \Rightarrow B = \begin{bmatrix} 1 - \cos T \\ \sin T \end{bmatrix}$$

$$x(k+1) = \begin{bmatrix} \cos T & \sin T \\ -\sin T & \cos T \end{bmatrix} x(k) + \begin{bmatrix} 1-\cos T \\ \sin T \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(k)$$

Matlab:

$$Sysc = ss(Ac, Bc, Cc, Dc);$$

$$Sysd = c2d(Sysc, T, 'zoh')$$

Solution of $x(k+1) = Ax(k)$

$$x(k+1) = Ax(k), \quad x(0) = x_0$$

$$x(1) = Ax(0) = Ax_0$$

$$x(2) = Ax(1) = A^2x_0$$

$$x(3) = Ax(2) = A^3x_0$$

$$x(k) = A^kx_0$$

Properties of A^k (compared with e^{At})

$$e^{At}\Big|_{t=0} = I \qquad A^k\Big|_{k=0} = I$$

$$\frac{d}{dt}e^{At} = Ae^{At} \qquad A^{k+1} = AA^k$$

$$e^{A(t+\tau)} = e^{At}e^{A\tau} \qquad A^{k+m} = A^k\,A^m$$

e^{At} is always nonsinguar

A^k is nonsingular if and if A is nonsingular

Complete Solution

$$x(k+1) = Ax(k) + Bu(k), \quad x(0) = x_0$$

$$x(1) = Ax(0) + Bu(0) = Ax_0 + Bu(0)$$

$$x(2) = Ax(1) + Bu(1) = A^2x_0 + ABu(0) + Bu(1)$$

$$\begin{aligned} x(3) &= Ax(2) + Bu(2) \\ &= A^3x_0 + A^2Bu(0) + ABu(1) + Bu(2) \end{aligned}$$

$$\begin{aligned} x(4) &= Ax(3) + Bu(3) \\ &= A^4x_0 + A^3Bu(0) + A^2Bu(1) + ABu(2) + Bu(3) \end{aligned}$$

$$\boxed{x(k) = A^kx_0 + \sum_{j=0}^{k-1} A^{k-j-1}Bu(j)}$$

The z-Transform Method

Review of the z-Transform

The z-transform of $f(k)$ is defined by

$$\hat{f}(z) = \mathcal{Z}\{f(k)\} = \sum_{k=0}^{\infty} z^{-k} f(k)$$

Properties

$$\mathcal{Z}\{f(k+1)\} = z[\hat{f}(z) - f(0)]$$

$$\mathcal{Z}\{f(k+\ell)\} = z^{\ell}\left[\hat{f}(z) - \sum_{i=0}^{\ell-1} z^{-i} f(i)\right]$$

$$\mathcal{Z}\{f(k-\ell)\} = z^{-\ell}\left[\hat{f}(z) + \sum_{i=-\ell}^{-1} z^{-i} f(i)\right]$$

$$\mathcal{Z}\{a^k f(k)\} = \hat{f}(z/a)$$

$$\mathcal{Z}\left\{\sum_{i=0}^{\infty} f(i)g(k-i)\right\} = \hat{f}(z)\hat{g}(z)$$

$$f(0) = \lim_{z\to\infty} \hat{f}(z)$$

$$\lim_{k\to\infty} f(k) = \lim_{z\to 1}(z-1)\hat{f}(z)$$

provided all poles of $(z-1)\hat{f}(z)$ are inside the unit circle

Complete Solution

$$x(k+1) = Ax(k) + Bu(k), \quad x(0) = x_0$$

$$z[\hat{x}(z) - x_0] = A\hat{x}(z) + B\hat{u}(z)$$

$$(zI - A)\hat{x}(z) = zx_0 + B\hat{u}(z)$$

$$\hat{x}(z) = z(zI - A)^{-1}x_0 + (zI - A)^{-1}B\hat{u}(z)$$

$$A^k = \mathcal{Z}^{-1}\left\{z(zI - A)^{-1}\right\}$$

$$\hat{y}(z) = C\hat{x}(z) + D\hat{u}(z)$$

Transfer function $(\hat{y}(z) = H(z)\hat{u}(z))$

$$H(z) = C(zI - A)^{-1}B + D$$

State Transformations:

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

$$x(k) = Pz(k)$$

$$\begin{aligned} z(k+1) &= P^{-1}x(k+1) = P^{-1}Ax(k) + P^{-1}Bu(k) \\ &= P^{-1}APz(k) + P^{-1}Bu(k) \end{aligned}$$

$$y(k) = CPz(k) + Du(k)$$

$$\{A, B, C, D\} \rightarrow \boxed{x = Pz} \rightarrow \{P^{-1}AP, P^{-1}B, CP, D\}$$

Transfer functions are invariant to state transformations

Jordan Form: There is a nonsingular matrix P such that

$$P^{-1}AP = J \quad \Rightarrow \quad A = PJP^{-1}$$

$$A^k = \left(PJP^{-1}\right)\left(PJP^{-1}\right)\cdots\left(PJP^{-1}\right)$$

$$A^k = PJ^kP^{-1}$$

If A is diagonalizable

$$J = \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \Rightarrow J^k = \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix}$$

$$A^k = \sum_{i=1}^{n} W_i \lambda_i^k$$

If A is not diagonalizable

$$J = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & . & \\ & & & . \\ & & & & J_r \end{bmatrix}$$

$$J_i = \lambda_i, \quad \text{or } J_i = \begin{bmatrix} \lambda_i & 1 \\ 0 & \lambda_i \end{bmatrix}, \quad \text{or } J_i = \begin{bmatrix} \lambda_i & 1 & 0 \\ 0 & \lambda_i & 1 \\ 0 & 0 & \lambda_i \end{bmatrix}$$

$$J_i = \lambda_i I + N_i$$

$$J^k = \begin{bmatrix} J_1^k & & & \\ & J_2^k & & \\ & & \ddots & \\ & & & J_r^k \end{bmatrix}$$

$$J_i^k = (\lambda_i I + N_i)^k$$

$$N_i = 0 \;\Rightarrow\; J_i^k = \lambda_i^k$$

$$N_i = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \;\Rightarrow\; N_i^k = 0, \text{ for } k \geq 2$$

$$J_i^2 = (\lambda_i I + N_i)(\lambda_i I + N_i) = \lambda_i^2 I + 2\lambda_i N_i$$

$$J_i^3 = \lambda_i^3 I + 3\lambda_i^2 N_i$$

$$J_i^k = \lambda_i^k I + k\lambda_i^{k-1} N_i$$

In general,

$$J_i^k = \lambda_i^k I + k\lambda_i^{k-1} N_i + \frac{k(k-1)}{2!}\lambda_i^{k-2} N_i^2 + \cdots + k\lambda_i N_i^{k-1} + N_i^k$$

For J_i of dimension $\ell \times \ell$

$$N_i^k = 0, \quad \text{for } k \geq \ell$$

Internal Stability

Study the behavior of λ^k and $k^\alpha \lambda^k$ (for positive integer α)

$$\lambda = re^{j\theta} \Rightarrow \lambda^k = r^k e^{jk\theta}$$

$$\left|\lambda^k\right| = r^k$$

$$\left|\lambda^k\right| \text{ is bounded } \Leftrightarrow r \leq 1$$

$$\lim_{k\to\infty} \left|\lambda^k\right| = 0 \Leftrightarrow r < 1$$

$\left|k^\alpha \lambda^k\right|$ is unbounded if $r > 1$ or $r = 1$ and $\alpha > 0$

Theorem: The system $x(k+1) = Ax(k)$ is

- Stable if and only if

$$|\lambda_i| \leq 1, \quad \text{for } i = 1, 2, \ldots, n$$

and for every eigenvalue with $|\lambda_i| = 1$ and algebraic multiplicity $q_i \geq 2$,

$$\text{rank}(A - \lambda_i I) = n - q_i$$

- Asymptotically (or exponentially) stable if

$$|\lambda_i| < 1, \quad \text{for } i = 1, 2, \ldots, n$$

Theorem: $H(z)$ is BIBO stable if and only if each pole of every element of $H(z)$ has absolute value less than one

Thank You